

A counterexample to the entropy polynomial root conjecture

8 October 2026

Abstract

Wakhare's entropy polynomial root conjecture asserts that an explicit polynomial $p_{k,r}$ has exactly two roots in $(0, 1)$ for every pair of integers $k > r \geq 1$. We give a counterexample with the coprime pair $(k, r) = (11, 10)$. Five exact rational sign evaluations yield at least four distinct roots.

1 The conjecture and the counterexample

In Conjecture 2 of [1], the entropy polynomials are defined by

$$h_{k,r}(x) = \sum_{j=0}^{k-1} x^{rj} \sum_{v=0}^j \frac{(-1)^{j-v}}{v+1} \binom{rv+k}{k} \binom{k}{j-v}, \quad (1)$$

and

$$p_{k,r}(x) = \alpha_{k/r} k(1-x^r)^k h_{k,k}(x) - r(1-x^k)^k h_{k,r}(x), \quad (2)$$

where $\alpha_s > 0$ is the unique solution of $\alpha_s(1+\alpha_s)^{s-1} = 1$. The conjecture states that $p_{k,r}$ has exactly two real roots in $(0, 1)$, counting multiplicity.

Theorem 1. *The polynomial $p_{11,10}$ has a root in each of*

$$(1/5, 2/5), \quad (2/5, 3/5), \quad (3/5, 2/3), \quad (2/3, 4/5).$$

In particular, Conjecture 2 of [1] is false, even when k and r are required to be coprime.

Write $\alpha = \alpha_{11/10}$, so $\alpha^{10}(1+\alpha) = 1$. Define

$$\begin{aligned} P(t) &= 1 + 352705t + 60632419t^2 + 1227099358t^3 \\ &\quad + 6330005947t^4 + 10701243741t^5 + 6330005947t^6 \\ &\quad + 1227099358t^7 + 60632419t^8 + 352705t^9 + t^{10}, \\ Q(t) &= 11 + 1939817t + 289126442t^2 + 5380098482t^3 \\ &\quad + 25959010187t^4 + 41238382790t^5 + 22851341183t^6 \\ &\quad + 4098130058t^7 + 181139618t^8 + 831413t^9 - t^{10}. \end{aligned}$$

Evaluation of the finite sums in (1) gives exactly

$$h_{11,11}(x) = P(x^{11}), \quad h_{11,10}(x) = Q(x^{10})/11. \quad (3)$$

Both $P(t)$ and $Q(t)$ are positive for $0 < t < 1$: all coefficients of P are positive, while the constant and final terms of Q sum to $11 - t^{10} > 0$ and its other coefficients are positive.

2 Exact signs and the proof

The function $z^{10}(1+z)$ is strictly increasing on $z > 0$. The integer identities

$$\begin{aligned} 117^{10} \cdot 242 - 125^{11} &= -90074807210933370467 < 0, \\ 937^{10} \cdot 1937 - 1000^{11} &= 10474898608767871728104442364513 > 0 \end{aligned}$$

therefore establish

$$117/125 < \alpha < 937/1000. \quad (4)$$

For $0 < x < 1$, put

$$R(x) = \frac{10(1-x^{11})^{11}Q(x^{10})}{121(1-x^{10})^{11}P(x^{11})}. \quad (5)$$

The denominator is positive, and (2)–(3) give

$$p_{11,10}(x) = 11(1-x^{10})^{11}P(x^{11})(\alpha - R(x)).$$

Thus the sign of $p_{11,10}(x)$ is the sign of $\alpha - R(x)$. The strict rational comparisons below are certified by the positive integers in the appendix.

x	Exact bound on $R(x)$	$\text{sgn } p_{11,10}(x)$
1/5	$R(x) < 919/1000 < 117/125$	+
2/5	$R(x) > 1131/1000 > 937/1000$	−
3/5	$R(x) < 935/1000 < 117/125$	+
2/3	$R(x) > 938/1000 > 937/1000$	−
4/5	$R(x) < 928/1000 < 117/125$	+

Since $p_{11,10}$ is continuous and these five ordered points lie in $(0, 1)$, the intermediate value theorem gives a root in each of the four disjoint intervals in Theorem 1. The roots are distinct, proving the theorem. $\square$

Scope. The argument proves *at least four* roots, and makes no assertion that there are exactly four. It does not refute the associated binary-entropy inequality, Conjecture 1 of [1], proved for all real exponents greater than one by Ho [2]. It also does not resolve Frankl’s union-closed sets conjecture.

Reproduction. The accompanying `check_certificate.py` reconstructs (3) from (1), checks (4), and verifies all five strict comparisons using integer arithmetic only. It uses no floating-point arithmetic, computer algebra system, or numerical root finder. Full numerator and denominator data are included in `certificate.json`.

A Cleared denominator certificate

Write $P(t) = \sum_{j=0}^{10} p_j t^j$ and $Q(t) = \sum_{j=0}^{10} q_j t^j$. For integers $0 < a < b$, define

$$\begin{aligned} \mathcal{P}(a, b) &= \sum_{j=0}^{10} p_j a^{11j} b^{11(10-j)}, & \mathcal{Q}(a, b) &= \sum_{j=0}^{10} q_j a^{10j} b^{10(10-j)}, \\ N(a, b) &= 10(b^{11} - a^{11})^{11} \mathcal{Q}(a, b), & D(a, b) &= 121b(b^{10} - a^{10})^{11} \mathcal{P}(a, b). \end{aligned}$$

Then $N, D > 0$ and $R(a/b) = N(a, b)/D(a, b)$. The following are exact positive integer values. Each decimal integer is wrapped across lines; its lines are to be concatenated.

$$919D(1, 5) - 1000N(1, 5) =$$

46687482475003686731728494443175716091029147392029835848827061539678
77835881726361080687327974249063360575405536849747865170141728596492
75824805705028730880

$$1000N(2, 5) - 1131D(2, 5) =$$

42538194776507513458711885524826212893895489706184796164361512069476
99426927690709188274361013225559293120108639340014430569971253115788
9439015076159529241505

$$935D(3, 5) - 1000N(3, 5) =$$

52516379104498148655863274985954973433649161277407686847051045991991
85256340399027956110708767358049927938282779151210248609234143827247
884850356820630516531200

$$1000N(2, 3) - 938D(2, 3) =$$

88008107586949258139134527029175262251555372081468008081118660645776
7956959351946751016842608896901672363281250

$$928D(4, 5) - 1000N(4, 5) =$$

12136820388591534024573035987433025741626348214182265217990319570111
14963599358661651488660016203235762863004128345364061271715446024425
596394854034905941742192560

References

- [1] T. Wakhare, *Iterated entropy derivatives and binary entropy inequalities*, Journal of Approximation Theory **307** (2025), 106143. doi:10.1016/j.jat.2025.106143. arXiv:2312.14743v2.
- [2] B. S. Ho, *A generalization of Boppana's entropy inequality*, arXiv:2601.19327 (2026).